



Deep Generative Models

From 2D Manifolds to Image Synthesis

George Boufis, f3352504

Abstract

This report compares three families of deep generative models on synthetic two-dimensional data and MNIST image generation: Variational Autoencoders, Generative Adversarial Networks, and Denoising Diffusion Probabilistic Models. First, the models are evaluated on a two-dimensional Gaussian mixture arranged in a ring, where the goal is to assess whether each model can capture a multimodal distribution. The experiments are then extended to MNIST using a Convolutional VAE, a DCGAN, and a lightweight DDPM with a Micro U-Net denoising network. The results show different trade-offs between fidelity, diversity, training stability, and inference cost. On MNIST, the DCGAN produced the sharpest digit samples, while ConvVAE produced smoother and blurrier but more stable samples. The DDPM and DDIM experiments demonstrated the iterative denoising mechanism of diffusion models, but also highlighted their main computational drawback: sampling requires many sequential reverse steps, unlike VAEs and GANs, which generate images with a single forward pass. Overall, the experiments show that no model dominates in every aspect, instead, each architecture reflects a different trade-off between sample quality, optimization stability, and computational efficiency.

Introduction

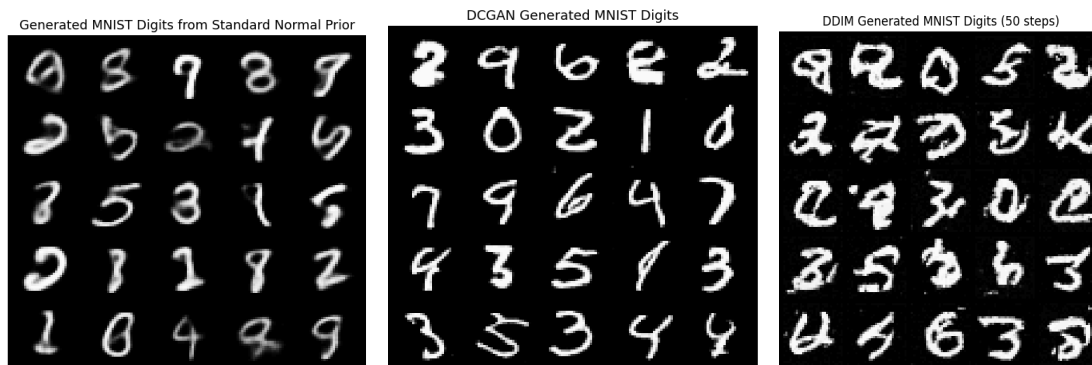
Deep generative models learn an underlying data distribution in order to generate new samples that resemble the training data. This work compares three main approaches: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and diffusion models. VAEs optimize the ELBO using a reconstruction term and KL regularization, which usually leads to stable training but can produce smoother or blurrier samples. GANs train a generator and a discriminator in an adversarial game, often producing sharper images but with less stable optimization. Diffusion models generate samples through an iterative denoising process, which can achieve high image quality but makes inference computationally expensive.

The experiments first evaluate the models on a two-dimensional Gaussian mixture distribution to study mode coverage and mode collapse. They are then scaled to MNIST using convolutional architectures. The MNIST comparison focuses on visual fidelity and diversity, VAE versus GAN training stability, and the inference cost of DDPMs compared with one-pass generators such as VAEs and GANs.

Fidelity vs Diversity

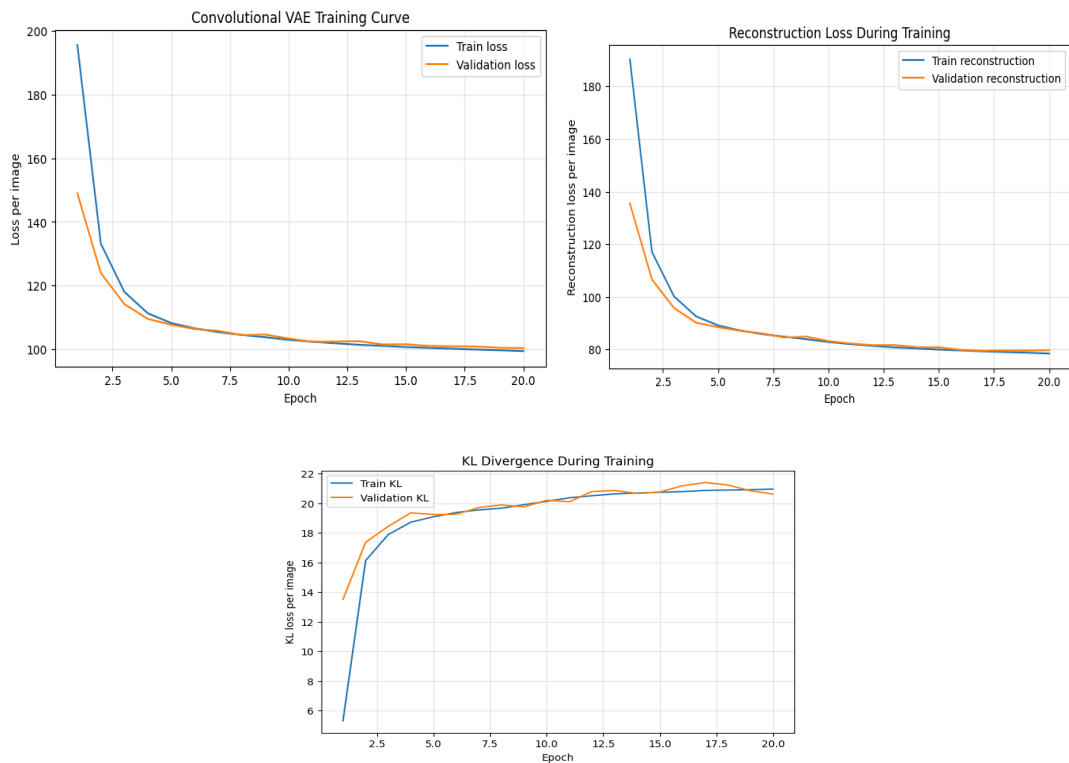
Based on the MNIST samples generated, the DCGAN produced the sharpest images overall, while the Convolutional VAE produced the blurriest images. However, blur and recognizability should be distinguished. The ConvVAE samples have softer and less well-defined strokes, but most of them still resemble valid digits. In contrast, the DDPM/DDIM samples are not necessarily blurry. Instead, several of them contain artifacts, unusual curves, broken strokes, or malformed digit structures. Therefore, in terms of visual sharpness, the DCGAN performs best and the ConvVAE is the blurriest, but in terms of digit recognizability the ConvVAE appears more consistent than the lightweight DDPM/DDIM implementation.

This behavior is consistent with the models' objectives. The VAE optimizes a variational lower bound using a reconstruction component and latent-space regularization, which often leads to smooth or averaged outputs. VAEs are also commonly reported to suffer from blurry generated samples and reconstructions. The DCGAN, on the other hand, is trained with an adversarial objective: the generator tries to fool a discriminator that distinguishes real from generated samples, which encourages sharper and more realistic local image structure. Finally, diffusion models can achieve high-quality image synthesis, but they rely on an iterative denoising process; with a small Micro-U-Net and limited training, the reverse process may still produce artifacts or malformed digits.



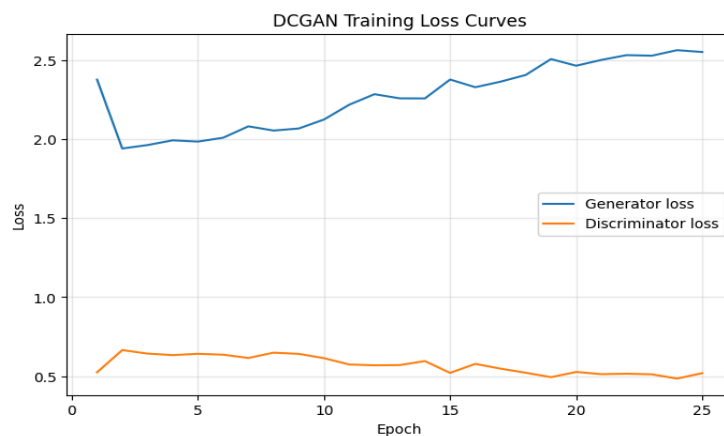
Training Dynamics

The ConvVAE showed stable training dynamics. Both the training and validation losses decreased smoothly across the 20 epochs, with the validation loss improving from 149.00 to 100.32. The reconstruction loss followed the same trend, indicating that the decoder progressively learned to reconstruct MNIST digits more accurately. The KL divergence increased during the early epochs and then stabilized around 20-21, suggesting that the latent space was being used without unstable growth or collapse. Overall, the VAE loss curves were smooth and directly interpretable, which is expected because the model optimizes a single ELBO-based objective.



In contrast, the DCGAN training curves were less directly interpretable. The discriminator loss fluctuated within a relatively narrow range, while the generator loss generally increased from around 1.94 to 2.55. This does not mean that the GAN failed, since the generated samples were visually sharp. However, it shows that GAN training was less stable and harder to interpret than VAE training. This is expected because the VAE optimizes a single ELBO-based objective, whereas the GAN is trained as a two-player adversarial minimax game between the generator and discriminator. As a result, improvements in one network can make the other network's task harder, leading to oscillating or non-monotonic loss curves.

Overall, the VAE was more stable and predictable during training, while the DCGAN produced sharper samples but had less stable adversarial training dynamics.



Compute Trade-offs

The primary computational drawback of DDPMs during inference is that sampling is iterative and sequential. Unlike VAEs and GANs, which can generate an image with a single decoder or generator forward pass, a DDPM starts from pure noise and repeatedly applies the denoising network over many reverse diffusion steps. The standard DDPM used $T = 250$ timesteps, meaning that generating one MNIST image required approximately 250 sequential denoising evaluations. This makes inference considerably slower than in the ConvVAE or DCGAN, where sampling only requires drawing a latent/noise vector and passing it once through the decoder or generator.

This drawback was also visible in the DDIM experiment. By using 50 DDIM sampling steps instead of the full 250 DDPM steps, it achieved a 5-fold reduction in the number of reverse steps. However, even 50 DDIM steps are still much more expensive than the single forward pass required by a VAE or GAN. Therefore, based on this experiment, the main compute trade-off of diffusion models is not training stability or sample quality, but slow inference due to the need for many sequential denoising steps.

Conclusion

Overall, the experiments highlight the different trade-offs between VAEs, GANs, and diffusion models. The ConvVAE trained the most smoothly and produced stable, recognizable but blurry MNIST digits. The DCGAN was less stable during training, as expected from its adversarial objective, but generated the sharpest samples. The DDPM/DDIM models demonstrated the strength of iterative denoising, but in this lightweight setup their samples contained artifacts and their inference was much slower than the one-pass generation of VAEs and GANs. Therefore, the results show that model choice depends on the desired trade-off: VAEs offer stability, GANs offer sharper visual samples, and diffusion models offer a powerful but computationally expensive generation framework.

References:

Kingma, D. P., & Welling, M. (2013, December 20). Auto-Encoding Variational

Bayes. *arXiv, abs/1312.6114*. <https://doi.org/10.48550/arXiv.1312.6114>

Kingma, D. P., & Welling, M. (2019, June 6). An Introduction to Variational

Autoencoders. *arXiv, abs/1906.02691*. <https://doi.org/10.1561/22000000056>

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014, June 10). Generative Adversarial Networks. *arXiv, abs/1406.2661*. <https://doi.org/10.48550/arXiv.1406.2661>

Radford, A., Metz, L., & Chintala, S. (2015, November 19). Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. *arXiv, abs/1511.06434*. <https://doi.org/10.48550/arXiv.1511.06434>

Ho, J. (2020, June 19). Denoising Diffusion Probabilistic Models. *arXiv, abs/2006.11239*. <https://doi.org/10.48550/arXiv.2006.11239>

Song, J., Meng, C., & Ermon, S. (2020, October 6). Denoising Diffusion Implicit Models. *arXiv, abs/2010.02502*. <https://doi.org/10.48550/arXiv.2010.02502>